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Elucidating supercritical fluid extraction of fucoxanthin from algae to enable the integrated biorefinery

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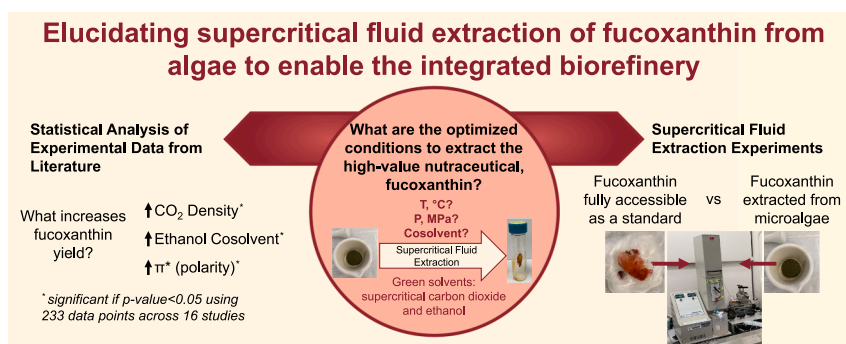
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HIGHLIGHTS

- Literature differs on fucoxanthin-optimized supercritical extraction conditions.
- Statistical analyses of literature data identify 3 predictors of fucoxanthin yield.
- Experiments establish how tunable solvent polarity drives fucoxanthin recovery.
- Extractions on fucoxanthin standard vs. algae show hindrance of the algae matrix.
- Fucoxanthin selectivity over chlorophyll is observed in supercritical CO₂ systems.

GRAPHICAL ABSTRACT



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ABSTRACT

The emerging nutraceutical, fucoxanthin, shows promise as a high-value product to enable the integrated biorefinery. Fucoxanthin can be extracted from algae through supercritical fluid extraction (SFE), but literature does not agree on optimal extraction conditions. Here, a statistical analysis of literature identifies supercritical carbon dioxide (scCO₂) density, ethanol cosolvent amount, and polarity as significant predictors of fucoxanthin yield. Novel SFE experiments are then performed using a fucoxanthin standard, describing its fundamental solubility. These experiments establish solvent system polarity as the key knob to tune fucoxanthin recovery from 0% to 100% and give specific operating conditions for targeted fucoxanthin extraction. Further experiments compare extractions on fucoxanthin standard with extractions from *Phaeodactylum tricornutum* microalgae to elucidate the effect of the algae matrix. Results show selectivity of fucoxanthin over chlorophyll in scCO₂ microalgae extractions that was not seen in extractions with ethanol, indicating a benefit of scCO₂ to design selective extraction schemes.

Abbreviations: SFE, supercritical fluid extraction; CO₂, carbon dioxide; scCO₂, supercritical carbon dioxide; FY, fucoxanthin yield; UHPLC, ultra-high pressure liquid chromatography; MS, mass spectrometry; vol%, volume.

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1. Introduction

Integrated biorefineries are seen as a key strategy in the transition towards a low carbon economy by reducing dependence on fossil fuels, utilizing renewable resources, and reducing greenhouse gas emissions. Unlike classical biorefineries that focused solely on producing biofuels with low sale prices, the goal of an integrated biorefinery is to maximize the value of the biomass feedstocks by efficiently utilizing all available components and creating multiple product streams at varying price points. For example, an integrated biorefinery might produce low-value, high-volume products (like biofuels) as well as high-value, low-volume biochemicals (like nutraceuticals) to become economically advantageous (Kwan et al., 2018; Wang et al., 2023).

Microalgae is a promising biomass feedstock for an integrated biorefinery (Kwan et al., 2018; Wang et al., 2023) as many species contain high-volume lipid fractions that can be converted into biodiesel along with low-volume carotenoids and xanthophylls that are of interest in nutraceutical, cosmetic, and pigment markets (Ballesteros-Vivas et al., 2022; Kwan et al., 2018). Research interest in fucoxanthin, a xanthophyll found in macro- and microalgae, has increased greatly in recent years (Khaw et al., 2021) due to its promising health benefits. As a nutraceutical, fucoxanthin is marketed primarily for weight loss, liver health, and as an antioxidant (Pajot et al., 2022). There is also evidence that it may help prevent or treat inflammation-related diseases such as cancer, obesity, liver disease, and damage to cardiovascular and nerve tissue (Abidov et al., 2010; Liu et al., 2020; Matsumoto et al., 2010; Pajot et al., 2022). As a cosmetic, fucoxanthin has been used as a safe and effective wrinkle reducer when applied topically due to its antioxidant characteristics (Foo et al., 2017; Kang et al., 2020; Pajot et al., 2022). A few manufacturers of algae-derived fucoxanthin products exist across the globe, including AlgaTech, AlgaHealth, Algae Health, and BrainPhyt. Both AlgaTech and BrainPhyt source their fucoxanthin from *Phaeodactylum tricornutum* microalgae to produce their dietary supplement pills containing 2–3 % fucoxanthin, while Algae Health sources from *Laminaria japonica* macroalgae. Due to these health benefits and uses, the global fucoxanthin market is estimated to increase to more than \$100 million by 2025 (Khaw et al., 2021). Importantly for the valorization of the integrated biorefinery, fucoxanthin has high value, priced at \$40,000–\$80,000 per kg depending on quality (Derwenskus et al., 2020).

Several methods exist to extract fucoxanthin from algae, with supercritical fluid extraction (SFE) emerging as a promising method for several reasons (Ashokkumar et al., 2023; Leong et al., 2022). First, SFE uses green solvents, supercritical carbon dioxide (scCO₂) with or without cosolvent modifiers such as ethanol. Both carbon dioxide and ethanol are considered green solvents since they have minimal to no environmental, health, and safety concerns, minimal to no emissions, are abundant in the environment, or can be renewably sourced (Soh and Eckelman, 2016). Especially advantageous for food, cosmetic, and medical applications, products extracted in these solvent systems are directly safe for human consumption since neither is regulated as a solvent by the United States Food and Drug Administration (United States Food and Drug Administration et al., 2018). This realizes cost savings by avoiding downstream processing to eliminate solvents with toxicity concerns. A key advantage of scCO₂ is its tunability through changes in operating temperature and pressure corresponding to changes in density, where higher density leads to greater solubility of polar compounds. In this way, changing the operating conditions can selectively target different fractions in algae biomass, as exemplified in the sequential extraction of triglycerides and astaxanthin in a 2-step scCO₂ extraction (Kwan et al., 2018). Fucoxanthin has been extracted with pressurized or scCO₂ from several species of macro- and microalgae (Table 1). Including ethanol as a cosolvent with pressurized carbon dioxide has generally been noted to increase the fucoxanthin yield. However, when aiming to apply the integrated biorefinery concept to the extraction of fucoxanthin from macro- and microalgae using SFE, a

review of experimental SFE literature shows inconsistencies in the reported “optimized” operating conditions, such that scientists or engineers who would want to design an extraction process do not have clear guidance on how to do so.

This work identifies the inconsistencies across literature—mainly due to differing types of algae species and differences across studies—but also statistically analyzes over 200 data points from 16 studies to identify variables that lead to increased fucoxanthin yield. In addition to identifying and analyzing results from literature, this work bridges the significant existing knowledge gap by conducting SFE experiments that describe the fundamental solubility of fucoxanthin in supercritical carbon dioxide, without any confounding factors (such as algae species) by using a pure fucoxanthin standard. Extractions of fucoxanthin standard are compared with extractions of fucoxanthin from the microalgae, *Phaeodactylum tricornutum*, to assess the effect of the algae matrix in maximizing recovery. This work further examines the effect of adding ethanol as a cosolvent and establishes solvent system polarity as the key knob to tune recovery of fucoxanthin; it provides specific operating conditions for targeted and tunable fucoxanthin extraction, while tracking selectivity of fucoxanthin over the co-product, chlorophyll. Overall, this work aims to elucidate the extraction of the high-value nutraceutical, fucoxanthin, with scCO₂ and ethanol as a cosolvent to advance the design and implementation of integrated biorefinery processes.

2. Materials and methods

2.1. Literature review and statistical analysis

A search of relevant literature on fucoxanthin extractions using scCO₂ was conducted using Web of Science (<https://www.webofscience.com>). A Web of Science search of fucoxanthin (All Fields) and supercritical (All Fields) generated 62 results on 1/24/2023. This search included the entire Web of Science database going back to the year 1900, with the earliest result from this search being from 2008. Each of the 62 papers was reviewed for relevancy and 16 papers were identified as experimental papers that reported on the optimization of fucoxanthin extraction (Table 1). Papers were included if fucoxanthin content was specifically reported, extractions used supercritical carbon dioxide or carbon dioxide plus ethanol (and in two cases, carbon dioxide plus an additional oil), and if an optimization was completed. Experimental studies that reported a single carbon dioxide extraction condition were not included.

Data from the 16 studies (Table 1) were compiled and relevant calculations were completed to relate fucoxanthin yield and recovery (Equation (1) and (2)) to the extraction conditions.

$$\text{Fucoxanthin yield (\%)} = \frac{\text{mg fucoxanthin}}{\text{g algae}} \times 100\% \quad (1)$$

$$\begin{aligned} \text{Fucoxanthin recovery (\%)} \\ = \frac{\text{mg fucoxanthin}}{\text{mg fucoxanthin extracted in conventional extraction}} \times 100\% \end{aligned} \quad (2)$$

After reviewing the data, fucoxanthin mass recovery was determined to be an inaccurate comparison between studies, since the conventional extractions that determine total amount of fucoxanthin available were not consistent across studies. Therefore, the literature comparison analysis will examine only fucoxanthin yield.

To compare the studies more easily, fucoxanthin yield was normalized using Equation (3):

$$\text{Normalized FY} = \frac{\text{FY at specific condition} - \text{minimum FY}}{\text{Maximum FY} - \text{minimum FY}} \quad (3)$$

Where FY is fucoxanthin yield, maximum FY is the highest yield obtained in an extraction with scCO₂ or scCO₂-ethanol cosolvent system in

the respective study, and minimum FY is the lowest yield obtained in an extraction with scCO₂ or scCO₂-ethanol cosolvent system in the respective study. Minimum FY to maximum FY describes the range of yields achieved in a study using scCO₂ or a scCO₂-ethanol cosolvent system and was used to normalize the data within each study so that the optimized condition was easier to identify across the comparison of

studies. Simple linear regressions where the response variable is related to a single predictor variable for each study were completed to weigh the effect of operating conditions to fucoxanthin yield. Response variables included operating temperature and pressure, carbon dioxide solvent density, volume percent of ethanol cosolvent, and Kamlet Taft parameters, α , β , and π^* . In evaluating the regression models, a p-value of 0.05

Table 1

Studies that report optimization of fucoxanthin from algae using supercritical or pressurized carbon dioxide (CO₂) extraction methods, with or without ethanol (CO₂ + EtOH) as a cosolvent, in order of most recent.

Ref	Algae Species (Macro/micro)	Solvent	Variables	Levels	Reported "optimized" condition
(Yin et al., 2022)	<i>Undaria pinnatifida</i> stem (Macro)	CO ₂ + EtOH	Temperature Pressure CO ₂ Flow Ethanol Cosolvent Extraction Time Particle Size	25, 40, 60 (°C) 1000, 3500, 6000 (PSI) 1, 2.5, 4 (mL/min) 1.25, 3.125, 5 (mL) 30, 135, 240 (min) 100, 450, 800 (µm)	50 °C, 41 MPa, 4 mL/min CO ₂ flow, 3.125 mL Ethanol, 100 µm
(Honda et al., 2022)	<i>Undaria pinnatifida</i> (Macro)	CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent Extraction Time	40, 80, 120, 160 (°C) 20, 30, 40 (MPa) 0.25, 0.5, 1 (mL/min) 15, 30, 60, 90, 120 (min)	160 °C, 30 MPa, 1 mL/min ethanol
(Ruiz-Domínguez et al., 2022)	<i>Phaeodactylum tricornutum</i> (Micro)	CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent	30, 50 (°C) 20, 30, 40 (MPa) 10, 20, 30, 40, 50 (vol% EtOH)	30 °C, 30 MPa, 40 vol% ethanol
(Ktari et al., 2021)	<i>Dictyopteris polypodioides</i> (Macro)	CO ₂	Temperature Pressure CO ₂ Flow	25, 40, 60 (°C) 30, 40, 50 (MPa) 15, 24 (mL/min)	60 °C, 50 MPa, 24 mL/min CO ₂
(Ktari et al., 2021)	<i>Isochrysis galbana</i> (Micro)	CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent	40, 50, 60 (°C) 20, 30, 40 (MPa) 0, 4, 8 (vol% EtOH)	60 °C, 30 MPa, 8 vol% ethanol
(Getachew et al., 2018)	<i>Saccharina japonica</i> (Macro)	CO ₂ + Coffee oil	Temperature Pressure Algae to Coffee oil Ratio	40, 45, 50 (°C) 20, 25, 30 (MPa) 25, 50, 75 (%)	40 °C, 30 MPa, 75 Algae to coffee ratio
(Jaswir et al., 2017)	<i>Sargassum binderi</i> (Macro)	CO ₂ + EtOH	Temperature Pressure Particle Size	40, 50, 60 (°C) 20, 25, 30 (MPa) 90, 500, 1000 (µm)	50 °C, 25 MPa, 500 µm
(Saravana et al., 2017)	<i>Saccharina japonica</i> (Macro)	CO ₂ , CO ₂ + EtOH, CO ₂ + Co-solvent	Temperature Pressure Ethanol Cosolvent Other Cosolvent (Soybean, Sunflower, Canola oils)	45, 50, 55 (°C) 20, 25, 30 (MPa) 0.5, 1.25, 2.0 (wt%) 0.5, 1.25, 2.0 (wt%)	55 °C, 30 MPa, 2.0 vol% EtOH 50 °C, 30 MPa, 2 vol% Sunflower oil
(Fabrowska et al., 2016)	<i>Cladophora glomerata</i> (Macro)	CO ₂ , CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent	40, 50, 60 (°C) 10, 20, 30 (MPa) 0, 7.5, 15 (mL/min)	40 °C, 30 MPa, 11.4 vol% EtOH
(Heffernan et al., 2016)	<i>Fucus serratus</i> (Macro)	CO ₂	Temperature Pressure	30, 40, 50 (°C) 150, 225, 300 (atm)	50 °C, 30.4 MPa
(Gilbert-López et al., 2015)	<i>Isochrysis galbana</i> (Micro)	CO ₂	Temperature Pressure	40, 50, 60 (°C) 10, 20, 30 (MPa)	50 °C, 30 MPa
(Balboa et al., 2015)	<i>Sargassum muticum</i> (Macro)	CO ₂	Pressure Moisture Content	10, 35 (MPa) 17, 24 (% moisture)	10 MPa, 17 % Moisture Content (45 °C)
(Conde et al., 2015)	<i>Sargassum muticum</i> (Macro)	CO ₂ , CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent	30, 40, 50, 60 (°C) 10, 15, 20, 25, 30 (MPa) 0, 0.5, 1, 2, 5, 10 (vol % EtOH)	50 °C, 20 MPa, 10 vol% EtOH
(Kanda et al., 2014)	<i>Undaria pinnatifida</i> (Macro)	CO ₂ , CO ₂ + EtOH	Temperature Pressure Ethanol Cosolvent Extraction Time	40, 50, 60, 70 (°C) 10, 20, 30, 40 (MPa) 0.05, 0.1, 0.3, 0.5 (mL/min) 30, 60, 90, 150, 210, 270 (min)	60 °C, 40 MPa, 0.1 mL/min EtOH (3.23 vol% EtOH)
(Quitain et al., 2013)	<i>Undaria pinnatifida</i> (Macro)	CO ₂	Temperature Pressure CO ₂ Flow Extraction Time	25, 40, 50, 60 (°C) 20, 30, 40 (MPa) 1.0, 2.0, 4.0 (mL/min) 30, 60, 120, 180 (min)	40 °C, 40 MPa, 4 mL/min CO ₂
(Roh et al., 2008)	<i>Undaria pinnatifida</i> (Macro)	CO ₂ + EtOH	Temperature Pressure Extraction Time	30, 40, 50, 60 (°C) 8, 10, 15, 20, 25, 30 (MPa) 10, 20, 30, 40, 50, 60 (min)	50 °C, 20 MPa

was used as the significance level metric.

2.2. Calculations of density and Kamlet-Taft parameters

Density of pure carbon dioxide and pure ethanol from temperature and pressure were calculated using the “RefProp” library in Aspen Plus Properties. These liquid phase densities were used to calculate mol percent flows of carbon dioxide and ethanol from mass and/or volume percent flows when necessary.

In order to relate fucoxanthin yield to properties of the solvent or solvent system during extraction, Kamlet-Taft parameters were used. Kamlet-Taft parameters are the most comprehensive and frequently used quantitative measure of solvent properties. Kamlet-Taft parameters are empirical measures of a solvent's (or a solvent mixture's) relative properties as a hydrogen bond donor (α), hydrogen bond acceptor (β), and its polarity/polarizability (π^*) (Jessop et al., 2012; Kamlet et al., 1977; Kamlet and Taft, 1976; Taft and Kamlet, 1976). The parameters are determined with solvatochromism by comparing the position of maximum absorption in the UV/VIS spectrum to conventional dyes (probes) and solvents. Relative wavelength transitions, blue shifts or red shifts, are represented by Equation (4):

$$V_{\max} = V_0 + a\alpha + b\beta + s\pi^* \quad (4)$$

where $V_{\max}$, V_0 , a , b , and s are properties of the probe and α , β , and π^* are solvent properties (Jessop et al., 2012; Kamlet et al., 1977; Kamlet and Taft, 1976; Taft and Kamlet, 1976). Then a scale is established using a measurement in two reference solvents as a zero and one. For example, π^* values of zero to one are in reference to cyclohexane and dimethyl sulfoxide (DMSO) respectively (Jessop et al., 2012). Kamlet-Taft parameters can be used to compare the properties of different solvents as well as single solvents and mixtures of solvents at different conditions (like in this work with carbon dioxide and carbon dioxide with ethanol cosolvent at different temperature and pressures). Kamlet-Taft parameters are measured with several specific dyes that can result in shifted α , β , and π^* values, and are often reported as an average of several dyes. Jessop et al. (2012) cover solvatochromic data, including Kamlet-Taft parameters, in more detail in their *Green Chemistry* review (Jessop et al., 2012).

Kamlet-Taft parameters were included from several sources to relate extraction conditions of temperature, pressure, and amount of ethanol cosolvent to π^* , α and β values. For supercritical carbon dioxide, π^* values at varying temperature and pressure from 5 studies (Duereh et al., 2020; Jessop et al., 2012; Kwan, 2018; Kwan et al., 2018; Sigman et al., 1985) were plotted versus corresponding carbon dioxide density. Trendlines were plotted for N, N-dimethyl-4-nitroaniline dye, 4-nitroanisole dye, and an average including all points with both dyes (see supplementary material). For comparison between supercritical carbon dioxide studies, the average π^* calculated from the linear relationship with density was used. For carbon dioxide with ethanol cosolvent, Kamlet-Taft values were available from two studies (Duereh et al., 2020; Skrebets et al., 2021). For ethanol cosolvent amounts, 0–10 mol % ethanol, π^* values were interpolated from Duereh et al. (2020) measurements for temperatures, 40–80 °C, and pressures, 1450–2900 PSI. For ethanol cosolvent amounts, 10–100 mol % ethanol, π^* , α and β values were interpolated from Skrebets et al. (2021) measurements for temperatures, 22–85 °C, and pressures, 2205–4410 PSI. Due to the limited set of temperatures and pressures where Kamlet-Taft parameters were measured, some extraction data points in the literature review could not be related to Kamlet-Taft values and were therefore excluded from analysis. Table 2 indicates the number of data points and number of studies included in each statistical analysis based on the data availability.

2.3. Experimental materials and methods

2.3.1. Materials

Bone dry carbon dioxide in gas cylinders equipped with siphon tubes (CD BD200S) and ultra-high purity nitrogen gas (NI UHP300) were obtained from AirGas, Inc. Chemical reagent grade, 200 proof (undistilled) ethanol was from Decon Labs. Fucoxanthin standard ($\geq 95\%$ purity, all-trans fucoxanthin, F6932) and chlorophyll *a* standard (quality level 200, powder from *Anacystis nidulans* algae, C6144) were sourced from MilliporeSigma. Glass wool (Catalog No: 3350) was purchased from Ohio Valley Specialty Company, USA. Water, isopropanol, acetonitrile were all liquid chromatography mass spectrometry grade solvents with 0.1 % formic acid either already in the solvent or added when preparing solvent mixtures.

2.3.2. Microalgae cultivation and harvesting

Phaeodactylum tricornutum microalgae was grown at the National Oceanic and Atmospheric Administration Northeast Fisheries Science Center Milford Laboratory. Growing conditions and harvesting methods were adapted from Ukeles (1973). In brief, *P. tricornutum* microalgae cultures were initially scaled by maintaining cultures under 12 h light, then 12 h darkness at 18 °C for the initial stage of growth. The cultures were then transferred into a larger flask and placed in a 20 °C room with 24-hour light for the next stage of growth. Cultures were then introduced into 19 L (5 gallon) glass carboys and maintained at 12 L scale with continuous air and carbon dioxide sparged through the culture under fluorescent lighting in a 20 °C room. The microalgae was grown with medium as described in Ukeles (1973), with the addition of silica every other harvest. *P. tricornutum* was harvested every 4–6 days by draining 6 L from the carboy and replacing with fresh media. The liquid was concentrated through centrifugation (Beckman, Model: J6 Refrigerated Floor Centrifuge, USA) and decanted as an algae slurry that was frozen at –80 °C and then freeze-dried (0.070 mBar, –47 °C) for approximately two weeks (Labconco, Model: FreeZone Freeze Dry Systems 7753524, USA). Once in a fully dried form, bags of dried algae were nitrogen capped, sealed, and stored in the freezer (–20 °C) until use. Before extraction, dried algae was ground into a fine powder for five minutes with a mortar and pestle. Image analysis of the ground powder indicated that particle size was typically smaller than 50 μm (see supplementary material).

2.3.3. Conventional extraction of fucoxanthin and chlorophylls

Conventional ethanol extractions were carried out to determine fucoxanthin and total chlorophyll content from microalgae samples before and after pressurized extractions in the continuous flow extractor. The method was adapted from Sarada et al. (2006) using

Table 2

Number of studies and data points for various statistical analyses that report optimization of fucoxanthin from algae using supercritical or pressurized carbon dioxide (CO₂) extraction methods, with or without ethanol (CO₂ + Ethanol) or oil (CO₂ + Oil) as a cosolvent. π^* , α , β are Kamlet-Taft parameters indicating polarity/polarizability, acidity, and basicity, respectively.

	Number of Data Points	Number of Studies
Overall	233	16
Optimized Conditions	23	16
CO ₂	11	11
CO ₂ + Ethanol	10	10
CO ₂ + Oil	2	2
Statistical models		
<i>Solvent – Response variable; predictor variables</i>		
CO ₂ – fucoxanthin yield; Temperature, Pressure, CO ₂ density, π^*	62	9
CO ₂ + Ethanol – fucoxanthin yield; Temperature, Pressure, Ethanol (vol %), π^*	130	8
CO ₂ + Ethanol – fucoxanthin yield; α , β	47	6

ethanol as a solvent, as ethanol allows higher recovery of fucoxanthin compared to other organic solvents (Perez et al., 2019). 50 mg of lyophilized, ground algae and 10 mL ethanol were added to centrifuge tubes. The tubes were vortexed for 1 min with a tabletop vortex mixer and sat for 1 h. Tubes were then vortexed again for 1 min and centrifuged in an Eppendorf centrifuge (Model: Eppendorf 5804, USA) for 10 min at 4200 rpm with a S-4-72 (164 mm) rotor. After centrifuging, the supernatant was decanted off the residual algae and stored in a vial. This ethanol extraction process was repeated three times total until the algae powder was colorless. The extract in ethanol was stored at -20°C until analysis.

2.3.4. Experimental set-up: supercritical fluid extractions

Continuous flow extractions were carried out in a 10 mL vessel in a Supercritical Fluid Technologies extractor (Model: SFT-100, USA) with built in temperature-controlled oven and heated outlet valves (see [supplementary material](#)). All experiments were at an oven temperature of 50°C . Carbon dioxide was pumped into the extractor by Teledyne Isco syringe pumps (Model: SyriXus 260X, USA) in continuous pressure, multi-pump mode set to the operating pressure. When ethanol was added as a cosolvent, an additional syringe pump (Teledyne Isco, Model: SyriXus 260X, USA) supplied a constant flow of ethanol. Total flow rate was controlled by adjusting the outlet valve until the SyriXus pump controller indicated the correct flow rate was achieved. For all experiments, the total flow rate was 2.7 ± 0.3 mL/min to match previously published work (Kwan et al., 2018). The outlet valve was heated to 70°C , which prevented freezing at the depressurization point from the Joule-Thompson effect but was below the boiling point of ethanol (78°C).

For each extraction, either $1500 \text{ mg} \pm 20 \text{ mg}$ of freeze-dried, ground *P. tricornutum* microalgae or $15 \text{ mg} \pm 2 \text{ mg}$ of fucoxanthin standard were loaded into the preheated extractor vessel with glass wool (to inhibit solvent channeling (Kwan et al., 2018)). 15 mg of fucoxanthin standard was chosen to simulate the magnitude of fucoxanthin present in 1500 mg of algae. The vessel was pressurized to the operating pressure and left to equilibrate for 30 min. After 30 min, the outlet valves were opened, ethanol flow was started (if using), and the total flow rate was set as indicated by the pump controller. The extraction proceeded for a total time of 2 h. The extraction product was collected in amber vials with glass wool in the caps (to prevent loss of product) sitting in an ice bath at the outlet of the extractor. At the end of the extraction, ethanol was introduced through a bypass to recover residual product remaining in the outlet line. Collected samples were diluted with ethanol and stored at -20°C until analysis. All extractions were completed in triplicate and reported with standard deviations of triplicates.

2.3.5. Conditions of supercritical fluid extractions

SFE operating conditions were selected throughout this study to appropriately compare solubility limitations of fucoxanthin in scCO_2 and to evaluate the effect of the algae matrix and the role of ethanol as a co-solvent on fucoxanthin recovery. In scCO_2 solvents, pressures of 13, 30, and 50 MPa at 50°C were selected to elucidate the limitations of fucoxanthin solubility and theoretical fit with the Chrastil model for solubility for pure standards and *P. tricornutum* microalgae. When ethanol was added as a cosolvent to pressurized CO_2 , extractions were held at 13 MPa (and 50°C) and contained set amounts of ethanol (5 %, 10 %, 40 %, 100 % ethanol by volume). These conditions allow the evaluation of a wide range of solvent system polarities from π^* values of -0.21 to $+0.52$ (see [supplementary material](#)).

2.3.6. Quantification of fucoxanthin and total chlorophylls

Fucoxanthin was quantified by ultra-high pressure liquid chromatography (UHPLC) and mass spectrometry (MS) from Thermo Fisher Scientific (Model: Vanquish Flex UHPLC Systems, USA) equipped with an MS (Model: TSQ Quantis Triple Quadrupole MS, USA) in electrospray ionization mode. The column (Thermo Scientific, Model: Accucore C30,

USA) was 150 mm in length. The eluent system was 70 % acetonitrile and 30 % water with 0.1 % formic acid for mobile phase A and 80 % isopropanol and 20 % acetonitrile with 0.1 % formic acid for mobile phase B. The solvent gradient started at 20 % B, 80 % A, then ramped to 90 % B over two minutes, then ramped to 98 % B over two minutes, held at 98 % B for two minutes, ramped to 100 % B over one minute, held at 100 % B for one minute, and then dropped back to 20 % B for two minutes to end the 10-minute total run. Identification, optimization, and calibration of fucoxanthin for UHPLC-MS analysis was conducted with 95 % all-trans fucoxanthin standard.

Total chlorophyll was measured spectrophotometrically (Wang et al., 2018) with an Agilent Technologies UV-Vis (Model: Cary 8454, USA) and quantified at 663 nm based on a calibration curve of chlorophyll *a* using Equation (5).

$$\text{Concentration} \left[\frac{\mu\text{g}}{\text{mL}} \right] = 11.7 \pm 0.7 \left[\frac{\mu\text{g}}{\text{mL}} \right] * A \quad (5)$$

where concentration is of total chlorophylls and A is absorbance at 663 nm. A calibration curve using chlorophyll *a* was created for each day of UV-Vis analysis and the slope value in Equation (5) is representative of the average plus or minus the 95 % confidence interval across 6 days of analysis.

3. Results and discussion

3.1. Statistical analysis of literature

When examining the optimized conditions for fucoxanthin extraction reported in literature (where “optimized” is defined by the study in terms of yield, purity, or recovery), the reported temperature and pressure set points vary considerably, with no clear “best” condition, although 50°C and 30 MPa is the most commonly reported optimized condition (Fig. 1a). One hypothesis for these discrepancies was that studies looked at only a limited range of conditions when an improved condition could have been obtained outside the range. However, for extractions in scCO_2 , the optimized condition was not always at the limit of the tested range (Fig. 1b). Out of the 11 studies that optimized for scCO_2 , 9 reported the optimized condition at the highest pressure examined, while only 4 reported optimal conditions at the highest temperature condition; and 6 were optimized at highest density condition. In addition, less than half of the studies that added ethanol cosolvents determined optimal conditions using the highest concentration of ethanol (4 out of 9 studies). These differences across studies can be attributed to how “optimized” was defined by the study, variation in algae species (both macro- and microalgae, with differing fucoxanthin content), and over-simplification of the comparison (e.g., comparing just temperature or pressure instead of density). These differences are important to acknowledge and understand when a scientist or engineer is designing their own fucoxanthin SFE system, as simply trying to replicate an extraction from these studies may not result in a truly optimized system for their own algae type or product specifications.

Since the literature did not converge on an optimized operating condition from the aggregate data, the 233 total data points were analyzed within each of the 16 studies individually to understand if significant predictors of fucoxanthin yield existed across studies and which variables are most strongly correlated with increased fucoxanthin yield. In linear regression models, single predictors were related to normalized fucoxanthin yield (Table 2). Single predictors included reported experimental operating conditions like temperature, pressure, and amount of ethanol cosolvent, that were also represented as carbon dioxide (CO_2) solvent density (from temperature and pressure), volume percent ethanol cosolvent (from reported volume percent or calculated from flow rate or mol percent), and Kamlet-Taft parameters, α , β , and π^* . Kamlet-Taft parameters indicate the relative acidity (α), basicity (β), and polarity (π^*) that the solvent or solvent system is exhibiting at the set operating conditions of temperature, pressure, and amount of ethanol

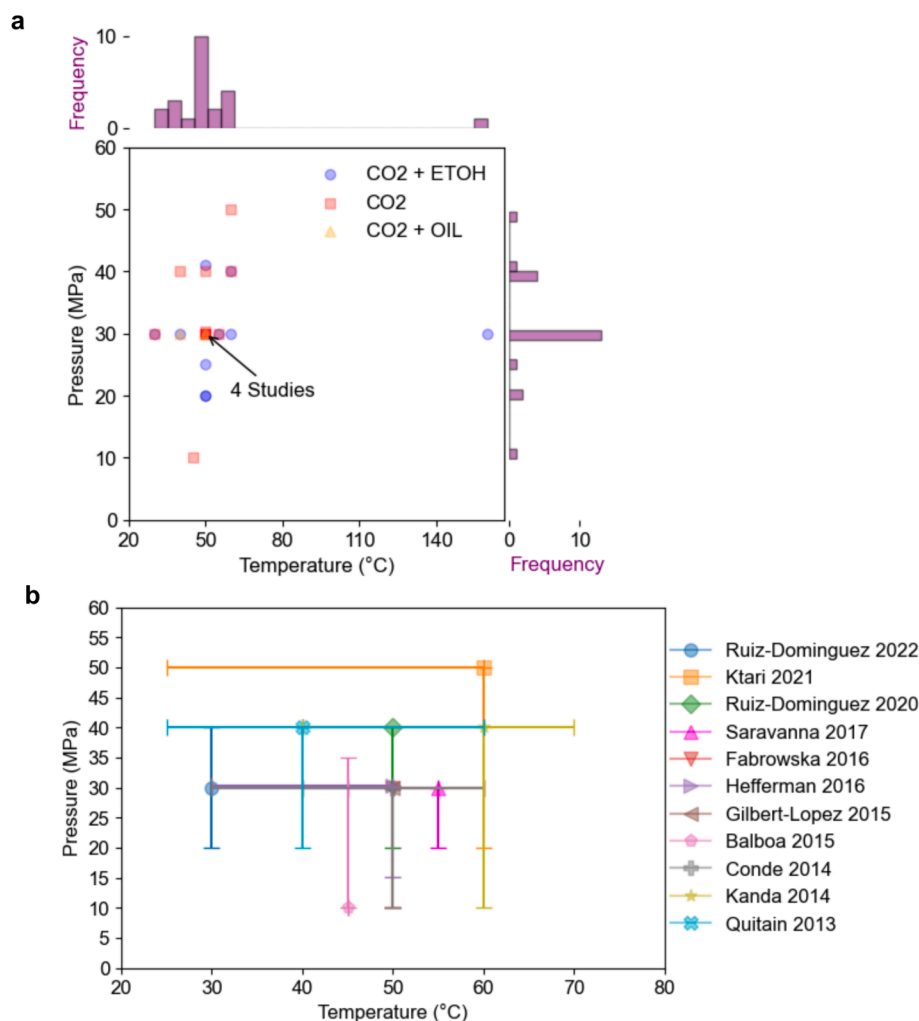


Fig. 1. (a) Optimized conditions of fucoxanthin extraction reported in literature using supercritical or pressurized carbon dioxide (CO₂) extraction methods, with or without ethanol (CO₂ + EtOH) or oil (CO₂ + Oil) as a cosolvent (Balboa et al., 2015; Conde et al., 2015; Fabrowska et al., 2016; Getachew et al., 2018; Gilbert-López et al., 2015; Hefferman et al., 2016; Honda et al., 2022; Jaswir et al., 2017; Kanda et al., 2014; Ktari et al., 2021; Quitain et al., 2013; Roh et al., 2008; Ruiz-Domínguez et al., 2022, 2020; Saravanna et al., 2017; Yin et al., 2022). (b) Optimized conditions of fucoxanthin extraction in supercritical carbon dioxide with the data point representing the reported optimized pressure and temperature condition and the bar as the range of conditions explored in the study.

cosolvent.

In the linear regression statistical models of data points from carbon dioxide extraction optimizations, π^* (polarity), density of carbon dioxide, and pressure, were often significant predictors (p -value < 0.05) of fucoxanthin yield (see [supplementary material](#)). In mixed carbon dioxide and ethanol systems, volume percent of ethanol cosolvent and π^* were significant predictors of fucoxanthin yield in the majority of examined studies (see [supplementary material](#)). Extraction temperature, α (acidity), and β (basicity) were rarely to never significant predictors of fucoxanthin yield (see [supplementary material](#)). Overall, the general conclusion is that higher carbon dioxide density, higher ethanol percentage, and higher π^* values correlate with increased fucoxanthin yield. Importantly, density is a function of temperature and pressure, and increased carbon dioxide density and volume percent of ethanol cosolvent are correlated with a more polar system, i.e. increased π^* .

While general conclusions can be drawn from the statistical analysis of literature, additional experiments were needed to clearly define the capabilities of scCO₂ extraction systems, to investigate the effect of the algae matrix on fucoxanthin recovery, and to understand the implications of adding ethanol as a cosolvent. We perform these experiments to elucidate the SFE extraction of fucoxanthin and present these results in the following sections.

3.2. Solubility of fucoxanthin in scCO₂ and the effect of the algae matrix

To first examine fucoxanthin extraction without any confounding factors, extractions were run with pure fucoxanthin standard in scCO₂ at different pressures to modulate carbon dioxide density and polarity to define maximum theoretical recoveries (Fig. 2). Since temperature is not a significant predictor of fucoxanthin yield, it was held constant at 50 °C, the mode optimized operating temperature (Fig. 1a). These pure standard extractions were compared with extractions from the microalgae, *P. tricornutum*, to assess the effect of the algae matrix on recovery of fucoxanthin.

In the extractions of pure fucoxanthin standard in scCO₂, recovery of fucoxanthin increases with increasing carbon dioxide density, matching the findings in the statistical analysis (Fig. 2a). The highest recovery achieved in scCO₂ was 30 % at 50 MPa (7250 PSI), which establishes that a recovery limit may exist in scCO₂ extractions, as even when fucoxanthin is entirely accessible as a pure standard full recovery is not achieved. While it is likely that higher densities of CO₂ can increase fucoxanthin recovery, extrapolation calculations reveal that carbon dioxide densities of 1090 kg/m³, corresponding to greater than 70 MPa (10,000 PSI) at 50 °C, would be required to reach 100 % recovery. Designing systems that approach or exceed 10,000 PSI can necessitate significant cost increases that hinder feasibility of the required

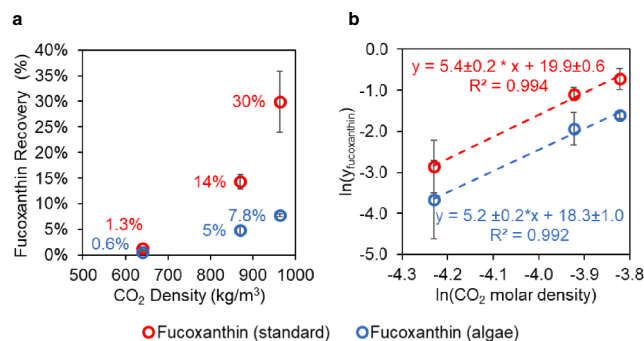


Fig. 2. (a) Fucoxanthin recovery related to the solvent density of supercritical carbon dioxide. (b) Chrastil model fits for fucoxanthin. Red circles indicate extractions on pure fucoxanthin standard; blue circles indicate extractions on *Phaeodactylum tricornutum* algae. Extractions were completed in triplicate at pressures of 13 MPa, 30 MPa, and 50 MPa. All extractions were at a temperature of 50 °C, a carbon dioxide flow rate of 2.7 mL/min, and for an extraction time of 2 h. Loadings were 15 mg of fucoxanthin standard or 1.5 g of *P. tricornutum* algae. Error bars represent standard deviation of triplicates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

extraction conditions.

Recovery of fucoxanthin is significantly lower when extracted from *P. tricornutum* microalgae than with the fucoxanthin standard, showing a clear effect of the algae matrix on the ability to recover fucoxanthin product (Fig. 2a).

The solubility of fucoxanthin in scCO₂ was further described using Chrastil's equation (Fig. 2b) which establishes a framework for solubility of solids and liquids in supercritical gases based on equilibrium thermodynamics (Chrastil, 1982). Chrastil's equation was simplified for constant temperatures as shown in Equation (6):

$$\ln(y_{\text{fucoxanthin}}) = k' \ln(\rho_{\text{CO}_2}) + b' \quad (6)$$

where k' describes the number of solvent molecules required to form a solvato complex with a single solute, and b' is related to the molecular weights of solute and solvent, as well as equilibrium heat and enthalpy of solvation (Chrastil, 1982; Kwan et al., 2016; Kwan and Zimmerman, 2019; Sparks et al., 2008). Continuous flow solubility for fucoxanthin in extractions with scCO₂ was calculated by Equation (7):

$$y_{\text{fucoxanthin}} = \frac{n_{\text{fucoxanthin}}}{n_{\text{CO}_2} + n_{\text{fucoxanthin}}} = \frac{n_{\text{fucoxanthin}}}{Q_{\text{CO}_2} \rho_{\text{CO}_2} t + n_{\text{fucoxanthin}}} \quad (7)$$

where $y_{\text{fucoxanthin}}$ is the mole fraction of fucoxanthin, $n_{\text{fucoxanthin}}$ is the total moles of fucoxanthin extracted, n_{CO_2} is the total moles of carbon dioxide delivered during the extraction, Q_{CO_2} is the flow rate of carbon dioxide, t is the total extraction time, and ρ_{CO_2} is the molar density of carbon dioxide at the extraction temperature and pressure.

Results from extractions of fucoxanthin standard and from *P. tricornutum* show the solubility of fucoxanthin in scCO₂ aligns well with the Chrastil model for solubility, which establishes a linear relationship on a log-log plot of the mole fraction of the solute and the density of scCO₂ for isothermal data (Fig. 2b). For both extractions of fucoxanthin from a pure standard and from microalgae, the linear fits have an R^2 value greater than 0.99. The slopes of the fits are statistically similar, indicating k' for both extractions with pure standard and with microalgae are similar. k' describes the number of solute molecules in a solvato complex so it logically follows that the values would be the same since the solute is fucoxanthin and the solvent is scCO₂ in both cases. For b' , the 95 % confidence intervals for the b' parameters scarcely overlap (19.9 ± 0.6 vs. 18.3 ± 1.0) and the larger error observed for extractions from the algae matrix are attributed to the low fucoxanthin recovery values obtained at the 13 MPa pressure conditions. Variation in the b'

parameter may be due to other solutes present in *P. tricornutum* algae that are not present in pure standard experiments, as Chrastil's model assumes no solute-to-solute interactions (Kwan et al., 2016; Kwan and Zimmerman, 2019), or by differences in enthalpy and entropy energies needed to reach thermodynamic equilibrium between fucoxanthin in a fully accessible form versus when pulled from the algae matrix (Kwan et al., 2016).

3.3. Elucidating the role of ethanol in CO₂-expanded ethanol cosolvent systems

Solvent system polarity can be modulated by either increasing density via pressure in scCO₂ or through the addition of a polar cosolvent, like ethanol. While other studies demonstrate the addition of ethanol as a cosolvent to increase yields of fucoxanthin from algae feedstocks (Conde et al., 2015; Honda et al., 2022; Ruiz-Domínguez et al., 2020; Saravana et al., 2017; Yin et al., 2022), the role of ethanol as a cosolvent and a theoretical descriptor of how ethanol modulates fucoxanthin recovery in CO₂-expanded ethanol cosolvent systems has not yet been established.

Here, the extraction results show that fucoxanthin recovery from a fully accessible standard can be tuned along the same trend by increasing the polarity through the addition of ethanol as a cosolvent or by increasing density by increasing the pressure of scCO₂ when plotted in terms of the Kamlet-Taft parameter, π^* (an indicator of solvent system polarity where polarity increases as π^* increases) (Fig. 3a). This reinforces the statistical analysis of literature reported fucoxanthin extractions suggesting that polarity is the strongest predictor driving fucoxanthin solubility in scCO₂ and cosolvent systems with ethanol. Notably, the addition of ethanol as a cosolvent allows for full recovery of fucoxanthin standard (100 % recovery) with as little as 40 volume percent (vol%) ethanol (at $\pi^* = 0.22$) by increasing the polarity of the solvent system. This result further informs design of the biorefinery processes in that less ethanol cosolvent is required to obtain full fucoxanthin recoveries, since 40 vol% ethanol extractions can yield the same fucoxanthin amounts as 100 vol% ethanol extractions, allowing for significant savings (in terms of solvent usage and potential need for solvent evaporation) when using mixed carbon dioxide and ethanol systems.

3.4. Evidence of selectivity in scCO₂ extractions

While fucoxanthin yields follow a consistent trend (Fig. 3a) a clear difference between fucoxanthin selectivity over chlorophyll exists between scCO₂ and CO₂-expanded ethanol cosolvent systems (Fig. 3b). While scCO₂ systems selectively extracted fucoxanthin while leaving chlorophylls in the algae matrix, ethanol-containing extractions solubilized chlorophylls into a non-selective product containing fucoxanthin and chlorophyll. This difference was observed even for solvent systems with nearly identical π^* values. For example, comparing extractions in scCO₂ at 50 MPa (corresponding to a π^* value of 0.04) and extractions at 13 MPa with 10 % volume ethanol cosolvent (corresponding to a π^* value of 0.00) the color of the extract is a deep orange/brown oil versus a deep green in an ethanol solution (Fig. 3c). Quantitatively, both extractions yield similar fucoxanthin recoveries, 8–9 %, but the chlorophyll recoveries vary significantly, 1.2 % versus 21 %. This difference can be evaluated across the polarity range as a ratio of fucoxanthin recovery to chlorophyll recovery (Fig. 3b). At pressures of 30 MPa and 50 MPa where recovery and yield of fucoxanthin is moderate, large benefits in selectivity are observed as chlorophyll is left behind in the remaining microalgae biomass, resulting in a higher ratio of fucoxanthin in the product extract. In extractions with 10 %, 40 %, and pure ethanol, increased amounts of fucoxanthin are recovered until full recovery is achieved, however the overall product is non-selective, with similar recoveries of fucoxanthin to chlorophyll in the 40 vol% and 100 vol% ethanol extracts (Fig. 3b).

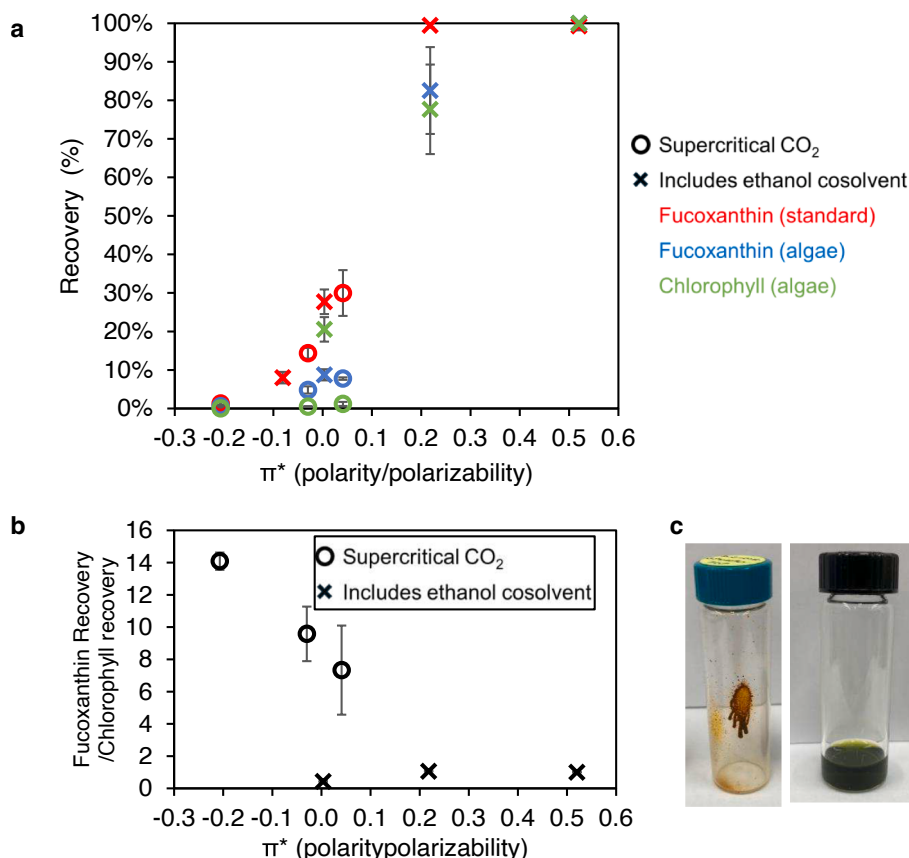


Fig. 3. (a) Fucoxanthin and chlorophyll recovery related to π^* (Kamlet-Taft parameter representing polarity/polarizability, where increasing π^* represents increasing polarity). O indicate extractions with supercritical carbon dioxide (scCO₂) (at pressures 13, 30, and 50 MPa). X indicate extractions with set amounts of ethanol (5 %, 10 %, 40 %, 100 % ethanol by volume, respectively) added as a cosolvent to pressurized carbon dioxide (CO₂) at 13 MPa. Extractions were completed in triplicate at temperature of 50 °C, a total flow rate of 2.7 mL/min, a fucoxanthin loading of 15 mg or algae loading of 1.5 g, and for an extraction time of 2 h. Error bars represent standard deviation of triplicates. (b) Ratio of fucoxanthin recovery to chlorophyll recovery where the dotted line indicates the same recovery of fucoxanthin and chlorophyll and points above the line indicate selectivity of fucoxanthin over chlorophyll. Error bars represent standard deviation of triplicates. (c) Photos of real extracts at similar π^* values (0.04 and 0.00, respectively), but (left) in scCO₂ versus (right) CO₂ with 10 % ethanol cosolvent, where the color visually shows selectivity of fucoxanthin over chlorophyll in scCO₂ extractions only.

The selectivity benefit in scCO₂ could be tied to carbon dioxide's gas-like and liquid-like properties in its supercritical state, with a similar diffusivity to liquids, but having lower viscosity, similar to gases (Uwina and Waśkiewicz, 2020). While supercritical CO₂ exhibits high diffusivity as a solvent, it is considered to be mild in solvent strength, viscosity, and permeability through the lipid bilayer relative to other organic solvents. These properties of organic solvents and their combined effect on solvent permeability assists in cell wall disruption. Comparatively, scCO₂ systems reportedly have difficulty overcoming analyte-matrix interactions for certain applications (Lehotay and Schenck, 2000) and can leave the lipid bilayer of freeze-dried algae cell walls mostly intact (Obeid et al., 2018). Furthermore, extraction from cellular matrices can be diminished by the rigidity of the cell wall (Obeid et al., 2018), which necessitates an organic solvent modifier to increase the polarity range to overcome retention of the matrix (Lehotay and Schenck, 2000). In a diatom like *P. tricornutum*, fucoxanthin is bound within a complex cellular structure called the fucoxanthin-chlorophyll protein complex, which is responsible for the light harvesting and photoprotective capacities of the cell (Wang et al., 2018). It is possible that the scCO₂ extraction only allows solubilization and release of the fucoxanthin pigment, while addition of an organic cosolvent like ethanol contributes to the degradation of the overall cell wall structure, as well as the fucoxanthin-chlorophyll protein complex, leading to non-selective recoveries of both fucoxanthin and chlorophyll.

4. Conclusions

This work provides fundamental knowledge that guides the design of biorefinery processes to isolate fucoxanthin from algae. Inconsistent literature on fucoxanthin extractions was clarified by a statistical analysis that identified scCO₂ density and amount of ethanol cosolvent (correlated to solvent polarity) as primary predictors of fucoxanthin yield. This work experimentally elucidates solubility and selectivity of fucoxanthin in scCO₂, the role of polarity tuning fucoxanthin recoveries in both scCO₂ and CO₂-ethanol cosolvent systems, and the effect of the microalgae matrix on fucoxanthin recovery. With this information, high-value fucoxanthin products can be produced synergistically with co-products, as intended by an integrated biorefinery.

CRediT authorship contribution statement

Mary Kate M. Lane: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Elise B. Gilcher:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Melissa M. Ahrens-Viquez:** Writing – original draft, Methodology, Investigation. **Rachel S. Pontious:** Writing – original draft, Methodology, Investigation. **Nora E. Wyrzten:** Investigation. **Julie B. Zimmerman:** Writing – review & editing, Visualization, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Julie B. Zimmerman reports financial support was provided by Gordon and Betty Moore Foundation. Mary Kate M. Lane reports financial support was provided by National Science Foundation Division of Graduate Education. Mary Kate M. Lane reports equipment, drugs, or supplies was provided by NOAA Fisheries Milford Laboratory. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2024.131036>.

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